GRECO: Geometric Registration of Electron Cryotomography Observations

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Abstract

Tomographic reconstruction is challenging because of limited angular coverage, noise, specimen motion and radiation damage. Here, I present GRECO, Geometric Registration of Electron Cryotomography Observations, an algorithm that uses local specimen structure to fit a shared three-dimensional geometry. Starting from central views, it selects landmarks from a working reconstruction and progressively adds neighbouring views. Near zero tilt, uncertainty in landmark depth has less effect on the predicted image position, allowing provisional landmarks to guide the first steps. Each landmark contributes through views with useful local information. For each observation, GRECO reconstructs a local reference from the other supporting views and compensates for the directional blur produced by those viewing angles. A joint fit updates image translations and landmark positions, linking overlapping angular tracks across the series. GRECO produced reconstructions suitable for subtomogram averaging, detection of small macromolecular complexes and detailed membrane segmentation across specimens ranging from vesicles with fiducials and a fiducial-free mixture to an intact near-minimal cell and cellular lamellae imaged in situ. The Greco software combines this registration algorithm with fast preprocessing, reconstruction and TinyCARE denoising, providing a practical workflow for studying molecular structure and cellular organization.

Introduction

Electron cryotomography (cryo-ET) reveals the three-dimensional organization of biological material, from isolated complexes to the crowded interior of cells. Images collected at different tilt angles are combined into a tomogram, where molecular structures can be studied together with their surroundings (Kelley et al., 2026). Repeated particles can then be aligned and averaged to recover finer structural detail (Zivanov et al., 2022). These analyses depend on accurate alignment of the images within a common three-dimensional geometry.

Accurate alignment is difficult because the electron dose is shared across the series and the accessible angular range is limited. The images are noisy, and structures at different depths overlap in projection. Their relative positions and appearance change as the specimen is tilted (Chen et al., 2019). Gold fiducials provide strong features for tracking and fitting the geometry (Mastronarde & Held, 2017), but suitable markers are not always present or visible across the series. Marker-free methods use specimen information through local feature tracks, correlation between views, or comparison with projections from a working reconstruction (Sorzano et al., 2009; Winkler & Taylor, 2006; Zheng et al., 2022). These methods have made automated reconstruction practical for many specimens.

A central challenge is to connect local information that remains useful over different parts of the angular range. A feature can provide a clear positional cue in several views and become difficult to distinguish in others. Previous work has shown how short landmark tracks can be combined to estimate the geometry (Sorzano et al., 2009), and how a working reconstruction can guide landmark localization and image alignment (Chen et al., 2019). However, a local match can still be ambiguous when structures overlap or the working volume is poorly constrained. A reference reconstructed from a limited set of views also carries directional blur. These considerations motivate an approach that connects the available observations, the construction of each local reference and the fit of a shared geometry.

Here, I introduce GRECO, Geometric Registration of Electron Cryotomography Observations. The algorithm selects three-dimensional landmarks from a working volume and

uses their local image patches to refine the geometry. A landmark only needs local structure that gives positional information, without requiring a molecular identity or an added marker. Selection searches columns through the volume from several viewing directions, spreading candidates across the image plane. Each landmark contributes through the views in which its surrounding structure can be used for registration. To measure a local correction, GRECO reconstructs a reference from the other supporting views, excluding the image being aligned. It compensates for the directional blur produced by that particular set of angles. The local corrections are then fitted jointly: landmarks in the same image share one image translation, and observations of the same landmark share one three-dimensional position.

GRECO starts from a small group of central views and progressively adds neighbouring views on both angular sides. Near zero tilt, the predicted image position is less sensitive to uncertainty in landmark depth. Provisional landmarks can therefore guide local matching before their depth is accurately determined. Each accepted expansion adds angular information, and the landmark cloud is rebuilt from the updated reconstruction. Overlapping angular tracks connect the local measurements across the series, allowing alignment and landmark localization to develop together.

The algorithm is implemented in Greco, which combines fast preprocessing and reconstruction with TinyCARE, a compact denoising network, and tools for particle analysis and membrane segmentation. The workflow produced reconstructions of ER-derived vesicles, a fiducial-free mixture of macromolecules, an intact near-minimal cell and cellular lamellae imaged in situ. These examples show how the same registration strategy can use different sources of specimen contrast and produce volumes for studying both molecular structure and cellular organization.

Approach

GRECO — Geometric Registration of Electron Cryotomography Observations — uses the specimen itself to align a tilt series. The algorithm is implemented by the Greco optimizer within the Greco software. A working reconstruction supplies candidate landmarks. Local image measurements then guide the fit of one shared three-dimensional geometry.

A landmark is a three-dimensional position used for local registration. It has an associated image patch in each view where the surrounding structure gives useful positional information. A landmark can represent part of a molecule, a small group of structures or a fiducial. No molecular identity or added marker is required. The optimizer estimates a translation for each image and a position for each landmark.

Geometric motivation and the central views

A feature moves in the image when the specimen is tilted, even without acquisition error. Consider a tilt axis parallel to the image y direction. Let d be the feature's lateral offset from the axis and h its depth along the beam axis at zero tilt (Figure 1a). Its projected position is

$$u(\alpha) = d\cos(\alpha) - h\sin(\alpha).$$

The difference between its zero-tilt and tilted positions is

$$\Delta x(\alpha) = u(0) - u(\alpha) = d(1 - \cos(\alpha)) + h\sin(\alpha). \tag{1}$$

The contribution from d is the same at positive and negative angles, while the contribution from h changes sign. Different features therefore follow different projected paths. Recon-

struction must preserve these relative motions while correcting the additional translation of each image. Coarse pairwise alignment gives a starting point, but different structures may drive the match at different angles. The resulting shifts still need to agree with one reconstruction geometry.

GRECO starts from a small central seed: neighbouring, coarsely aligned views used to build the first working reconstruction. Near zero tilt,

$$\Delta x(\alpha) \approx h\alpha + d\frac{\alpha^2}{2},$$

with α in radians. For fixed d and h, both contributions become small as the angle approaches zero. This helps nearby central views retain similar local arrangements. The displacement still depends on the feature's position and the angular range, so coarse alignment remains necessary.

The seed covers only a narrow angular range. Its reconstruction is therefore poorly constrained in depth (Davison, 1983). A feature can be better localized in the image plane than along the beam axis. The provisional landmark can therefore have a large depth error even when its lateral position is reasonably well determined. Let δd and δh be the estimated coordinates minus the true coordinates. The error in the predicted image position is

$$\delta u(\alpha) = \delta d \cos(\alpha) - \delta h \sin(\alpha).$$

If the lateral position is well localized, the main contribution is $-\delta h \sin(\alpha)$ (Figure 1b). Near zero tilt, a large depth error can therefore give a much smaller image error. For example, a depth error of 16 voxels gives a projection error of about 2.5 pixels at 9° and 13.9 pixels at 60°, with equal voxel and detector sampling. These are geometric examples, not measured alignment errors. Lateral errors are not reduced in the same way: at zero tilt, $\delta u(0) = \delta d$.

This is the reason for using provisional landmarks from the seed. Their depth can be imprecise, yet their projections can fall close enough to the feature in nearby views for local matching. The positions are then refined as angular coverage grows. GRECO selects the middle of the usable angle-ordered series; the explanation above applies most directly when this seed is near zero tilt.

For the joint fit, a landmark has centred coordinates $\boldsymbol{r} = (z, y, x)$. With the axis along y, its projection into view t is

$$\pi_t(\boldsymbol{r}) = (y, x\cos(\alpha_t) - z\sin(\alpha_t)). \tag{2}$$

Here, x and z correspond to d and h. The supplied tilt angles and axis orientation remain fixed during the core fit. Image translations are additional unknowns shared by all landmarks in each view.

Landmarks and their supporting views

GRECO uses a structural saliency score to identify local density useful for registration. The score is computed at each voxel in the region used for landmark selection. A three-dimensional structure tensor summarizes the strength and directions of nearby intensity gradients, following the gradient-based description of local image structure (Harris & Stephens, 1988). Its smallest eigenvalue favours intensity changes in all spatial directions. GRECO combines this response with a measure of compact positive density above the local background. The two contributions are normalized separately and combined through their geometric mean to give the structural saliency score.

Columns organize the selection to distribute candidates across the image plane, looking through the volume from different directions. Simply taking the highest scores could concentrate candidates around a few strong structures. GRECO divides a projection plane into a regular grid. Each grid region defines a column through the volume along that viewing direction, and its valid point with the highest structural saliency score becomes a candidate. The search is repeated for the proposed viewing directions, and the candidates are combined into one three-dimensional landmark cloud. This distributes the evidence available to the fit across the usable field of view. It does not prevent all overlap: points separated in three dimensions can still project into the same image region.

Figure 1c shows why this selection can work before depth is well determined. The central seed reconstructs a simulated fiducial as a feature elongated along the beam axis at zero tilt. The structural saliency score selects a point on this density, but does not remove its depth uncertainty. Although the selected point is displaced mainly along h, its projection into the nearby +12° view remains close to the fiducial. The score supplies a starting position for registration; it is not a measure of localization accuracy. The same selection is used for local specimen structure.

An observation is the image patch around a landmark's predicted projection in one view. Its usefulness can change with angle as the structure changes appearance or overlaps with other density. Local contrast and detector coverage determine the landmark's initial angular support: the views eligible to contribute observations. Different landmarks can have different supports. An out-of-frame observation is treated as unknown; insufficient contrast inside the valid image is treated as unsupported.

Supported observations form angular tracks linked by one landmark position. A track can cover only part of the series, and a landmark can have several supported segments. Overlapping tracks connect different angular ranges (Figure 1d). Candidates at identical coordinates are combined. Redundant nearby candidates are removed while preserving the existing links between pairs of views. The landmark cloud is rebuilt between stages as the working reconstruction gains angular information.

Local predictions and shared geometry

The working volume proposes the locations; the image patches supply the local references. For landmark k and target view t, GRECO reconstructs a small reference volume from the other supporting views:

$$\tilde{V}_{k,-t} = m\mathcal{F}^{-1}\left[K_{\mathcal{S}_{k,-t}}\mathcal{F}\big(\mathrm{WBP}\{p_{k,u} : u \in \mathcal{S}_{k,-t}\}\big)\right]. \tag{3}$$

The patches $p_{k,u}$ have zero mean and unit standard deviation. The set $\mathcal{S}_{k,-t}$ contains the supporting views except t. Weighted backprojection (WBP) filters these patches and places them along their viewing directions. $\mathcal{F}$ is the Fourier transform. The fixed spherical mask m keeps the centre of the local volume and tapers its boundary, avoiding a sharp box edge during projection. Excluding the target prevents its image values from directly entering its own reference. The references still share views and depend on the current geometry, so they are not statistically independent.

K compensates for the directional blur caused by reconstruction from the supporting views. GRECO estimates this blur by projecting an ideal point, or impulse, and reconstructing it with the same views and WBP procedure used for the local reference, excluding the target view. This response describes how those views spread a point in three dimensions. Its Fourier transform is normalized, then a regularized inverse gives the correction (Orieux et al., 2010). Regularization and a gain limit prevent excessive amplification where the

views transfer little information. A smooth low-pass taper reduces the correction at high spatial frequencies. The correction depends on the supporting views of each reference and is separate from the microscope CTF.

The reference is projected into the target direction and compared with the observed patch using generalized cross-correlation. A bounded search, with subpixel refinement where the peak permits it, gives a local image correction $\boldsymbol{v}_{k,t}$. This is the shift to apply to the observed image to match the current reference. All usable corrections are fitted together:

$$\min_{\Delta \boldsymbol{s}, \Delta \boldsymbol{r}} \sum_{(k,t)\in\mathcal{O}} \sum_{a\in\{y,x\}} \rho_\tau\Big(\Delta s_{t,a} - \left(J_t \Delta \boldsymbol{r}_k\right)_a - v_{k,t,a}\Big), \quad \sum_k \Delta \boldsymbol{r}_k = \boldsymbol{0}. \tag{4}$$

Here, $\Delta \boldsymbol{s}_t$ is the translation update of image t, and $\Delta \boldsymbol{r}_k$ is the position update of landmark k. The matrix J_t projects a position update using Equation 2. The difference $\Delta \boldsymbol{s}_t - J_t \Delta \boldsymbol{r}_k$ is the correction relative to the updated landmark, which explains the sign in the fit. $\mathcal{O}$ contains the usable landmark–view pairs, and a indexes the image directions. The soft-L1 loss ρ_τ, with scale τ, reduces the influence of large residuals.

A common translation of all landmarks can be compensated by corresponding image translations. The zero-sum constraint removes this ambiguity from the updates; it does not determine the absolute specimen position relative to the mechanical axis. Schur elimination gives a smaller system for the image translations, followed by recovery of the landmark updates (Triggs et al., 2000).

Equation 4 links the local measurements through one shared geometry. Each image has one translation shared by its landmarks, and each landmark has one three-dimensional position shared by its views. The fit updates both together to explain the measured local corrections. Overlapping tracks connect the views: an image shift affects every landmark observed in that image, while a landmark update changes its projections across views. More landmarks with overlapping tracks add constraints to the shared fit. A wider angular range helps determine landmark depth.

Natural-feature tracks, reconstruction-guided alignment and joint geometric fitting have established precedents (Chen et al., 2019; Han et al., 2019; Sorzano et al., 2009; Winkler & Taylor, 2006). Progressive projection matching with target exclusion also appears in AreTomo (Zheng et al., 2022). GRECO combines column-based landmark selection, partial angular support and local references with support-specific angular compensation to supply measurements for the shared fit.

Progressive refinement

A stage refines a fixed set of views before further angular expansion. GRECO first determines which views support each landmark. It then repeats local reference construction, correlation and joint fitting using the updated geometry (Figure 1e). Correlation measures local corrections within this support.

An observation is used only if its image patch has finite values and sufficient contrast. The complete patch and search window must fit within the original image, accounting for in-plane rotation and accumulated image shifts. This keeps artificial image borders out of the correlation. Patches are sampled from the original preprocessed images using the current total transformation, avoiding repeated interpolation of transformed images.

A stage is accepted when image updates remain small for several consecutive passes and the geometric checks pass. GRECO then attempts neighbouring views. The accepted images provide a new working volume. An incoming view is first positioned using landmarks supported by the nearest accepted view. A fresh landmark set is selected from that working

volume, and support for the incoming view is inferred from nearby accepted views. The expanded set is then refined jointly. Newly accepted images enter the working reconstruction for the next stage. This adds angular information about landmark depth and further constraints on the shared image translations.

If initial matching fails on one angular side, the other side can advance. The blocked side can then be tried again with more accepted views. A failed joint stage stops expansion and preserves the last accepted geometry, when one is available. Additional rounds can rebuild the landmarks and refine the accepted range before final reconstruction. Small updates indicate numerical convergence; alignment accuracy requires separate validation.

The optional Smoothness setting adds measured relative image shifts between neighbouring views to the landmark fit. These constraints can guide the alignment when landmark measurements are weak. They do not force the translations to follow a prescribed smooth curve. Equation 4 describes the landmark part of the objective; Smoothness adds the neighbouring-view terms.

An optional geometry refinement further adjusts image translations and viewing directions. It uses either the reconstruction landmarks or particle positions from subtomogram averaging. Appendix I describes estimation of the common tilt-axis orientation from the central seed.

Results

Evaluation on public data

The evaluation used publicly available cryo-ET movies from four datasets: ER-derived vesicles, the CZII object-identification challenge, near-minimal cells, and *Chlamydomonas reinhardtii* imaged in situ. Table 1 summarizes the specimens, the number of tilt series and the final reconstruction binning. The examples cover isolated complexes, crowded mixtures and cellular specimens. Picking was used to prepare downstream analyses, rather than as a benchmark of detection accuracy.

Table 1
Specimens and data used in this study.

Dataset and specimen	Specimen context	Series	Pixel size (Å/px)	Final bin
ER-derived vesicles HEK293F cells	Membranes, ribosomes and gold fiducials	1	1.724	4
CZII challenge · DS-10440 Five macromolecular classes	Mixed specimen without fiducials	7	1.510	8
DS-10442 · RN-16210 *M. mycoides* JCVI-Syn3A	Intact, plunge-frozen near-minimal cell	1	1.540	8
EMPIAR-11830 *C. reinhardtii*	Cellular lamellae imaged in situ	27	1.910	8

ER-derived vesicles: following local features during alignment

This example uses a tilt series of vesicles derived from the endoplasmic reticulum (ER) of HEK293F cells. The raw movies are available from DataverseNL (Chaillet, 2024). The series contains 35 tilts and was acquired at 200 kV. The input axis angle was −88.7°. All 35 views were accepted by the optimizer, and the final tomogram was reconstructed at 6.896 Å per voxel.

The landmark projection near zero tilt contains many visible gold fiducials (Figure 2a). Their compact, strong contrast makes them useful registration features, so GRECO tends to select them. Other selected regions contain specimen density. Landmarks do not need to represent a particular object: biological structures and ice contamination can also provide useful image information. The optimizer selects positional information, without assigning a biological identity to each landmark.

Figure 2b follows the same region through five stages of a Greco optimizer run, from a seven-view seed spanning ±9° to the full 35-view range of ±51°. The region contains a fiducial, ribosomes and a membrane. The side view shows how the fiducial's long initial extension along the beam axis at zero tilt becomes more localized as angular coverage increases.

Template matching used an 80S ribosome template from EMD-2938 with a soft density mask. After classification, 169 ribosomes were retained. Local subtomogram refinement reported a resolution of 29.3 Å from its phase-randomization-corrected FSC. The same 169 refined particle placements form the ribosome annotation layer. Figure 2c combines a TinyCARE section with membrane surfaces in blue and ribosome averages in red. Each average is placed at the corresponding particle position and orientation.

CZII challenge: a mixed specimen

The seven experimental training series from the CZII challenge contain a crowded mixture of macromolecules (Peck et al., 2025). GRECO used all 31 images in each tilt series. The final sampling was 12.08 Å per voxel.

GRECO selected 1,080 three-dimensional landmarks in the reconstruction shown, without fiducial markers. Of these landmarks, 694 had support at −24° and 687 at +24° (Figure 3a). The two projections show density around these landmarks. Their surrounding regions overlap, so several landmarks can contribute to the same visible feature. Selection from different directions produces overlapping angular tracks. This redundancy helps constrain a consistent three-dimensional geometry across views.

Across the seven reconstructed tomograms, template matching detected 445 apoferritin particles, 161 β-galactosidase particles, 467 thyroglobulin particles, 332 cytosolic ribosomes and 92 PP7 virus-like particles. The templates came from EMDB entries EMD-41923, EMD-0153, EMD-24181, EMD-3883 and EMD-41917, respectively. After extraction and particle selection, the final averages used 445, 136, 355, 324 and 90 particles in the same order. These numbers describe the detected candidates and the particles used for averaging; they do not by themselves measure picking precision or recall.

Figure 3b combines a TinyCARE section from the same tomogram with the refined particle averages, after geometry refinement guided by ribosome positions from STA. The averages are placed at their refined positions and orientations. The perspective view includes particles over a range of depths around the displayed section, showing their distribution within the specimen. The five final averages form the colour legend (Figure 3c).

DS-10442: an intact near-minimal cell

Run RN-16210 belongs to the near-minimal-cell dataset DS-10442 (Hutchings et al., 2026). It contains 31 views of *Mycoplasma mycoides* JCVI-Syn3A, a near-minimal cell with a reduced genome of 493 genes (Breuer et al., 2019). The cells were plunge-frozen intact, without FIB thinning. This provides a whole-cell example alongside the mixed specimen and cellular lamellae examined here.

The source study describes this specimen as thicker and more crowded than its two-protein phantom. It estimated a median specimen thickness of about 260 nm for this dataset (Hutchings et al., 2026). This is a dataset statistic, not an individual thickness measurement for RN-16210. The crowded cellular volume makes this a useful test of alignment when many structures overlap in projection.

This series was difficult for GRECO to align. The run without Smoothness did not progress beyond the seven-view seed, spanning −9° to +9°. A later run with Smoothness set to 4 accepted all 31 views, spanning −45° to +45°.

The CryoET Data Portal provides an aligned, denoised and annotated tomogram for this run (Paraan et al., 2024). Figure 4a,b compares a corresponding section from the deposited denoised map with the Greco reconstruction after TinyCARE denoising. Both are displayed at 12.32 Å per pixel. More intracellular detail is visible in the Greco overview, and the intracellular densities are better defined in the paired enlargements. Multiple bacterial ribosomes are visible in the tomogram (Figure 4c).

EMPIAR-11830: an in situ ribosome reconstruction

The in situ analysis included 27 tilt series of *C. reinhardtii* cellular lamellae from EMPIAR-11830 (Kelley et al., 2026). Twenty-six series contain 41 images and one contains 38 images. The input pixel size was 1.91 Å and the nominal axis angle was −95°. This experiment extends the application of GRECO to in situ lamella data, where the structures of interest remain inside a crowded cellular environment.

Greco reconstructed all 27 tomograms. Figure 5a–c shows three examples after geometry refinement guided by ribosome positions from STA and TinyCARE denoising.

After classification, 11,976 particles contributed to the ribosome refinements. The analysis included local refinement of particle positions and orientations, CTF refinement and polishing. Figure 5d shows the whole-ribosome average, refined with a broad mask. Its reported phase-randomization-corrected FSC resolution is 6.36 Å. A separate refinement with a mask around the large subunit reported 5.38 Å. These Greco reconstructions allowed both three-dimensional ribosome averaging and segmentation of membranes with different shapes, including extended sheets and curved surfaces (Figure 5a–c).

Denoising with TinyCARE

TinyCARE and cryoCARE were trained on the same even/odd half-volumes from the ER-derived vesicle dataset. Figure 6 shows the same region in the noisy input, after 100 to 5,000 TinyCARE training steps, and after the default 20,000 cryoCARE steps. The local structures become easier to see as TinyCARE training progresses.

The Greco-denoised tomograms shown in this article used 3,000 TinyCARE steps, except for the labelled checkpoints in Figure 6. This setting kept training short while retaining visible detail. The comparison is visual and does not measure denoising accuracy.

On an NVIDIA L40S, TinyCARE took about 3.4 min for training and prediction, compared with 50.4 min for cryoCARE. The measured times for the same 960 × 960 × 480 voxel tomogram were:

Method · steps	Training (s)	Prediction (s)	Total (s)
TinyCARE · 3,000	185.2	20.9	206.1
cryoCARE · 20,000	2,973.0	48.4	3,021.3

These times describe the two processing settings used here, which differ in network size, numerical precision and number of training steps.

Methods

Movie preprocessing and CTF estimation

Greco applies the detector gain reference and corrects defective pixels in the movie frames. Motion correction estimates a two-dimensional translation for each frame before summing them into one image per tilt. EER decoding, defect correction and full-frame motion correction use routines adapted from MotionCor3, which is integrated in AreTomo3 (Peck et al., 2026).

The contrast transfer function (CTF) was estimated from averaged power spectra of 512 × 512-pixel tiles at native sampling, with 50% overlap. Defocus and astigmatism were fitted from coarse to fine, using neighbouring tilts to initialize weak fits. These routines were adapted from AreTomo3 (Peck et al., 2026; Zheng et al., 2022). Spatial CTF correction accounts for the change in defocus across a tilted specimen. Defocus handedness determines which side of the image has higher defocus for the supplied tilt geometry. Greco tests the two signs by rescaling the local spectra to a common defocus and comparing their agreement with the fitted CTF. The local spectra also refine two angular offsets, shared across the series, to describe the specimen-plane orientation. The selected sign and refined gradient are saved for image correction and particle-level defocus calculations.

The images used for registration were CTF-corrected with a regularized inverse (regularization 0.05) and Fourier-binned to bin 8. Even and odd movie-frame sums were processed in the same way. Their agreement in Fourier space provided a reliability weight that reduced the contribution of poorly supported frequencies in the combined image.

Processing used 200 kV for the ER-derived vesicle dataset and 300 kV for the other datasets, with a spherical aberration of 2.7 mm and amplitude contrast of 0.1. The exposure inputs were 1.91, 2.03, 3.87 and 3.5 electrons/Å^2 per tilt for the ER-derived vesicles, CZII, DS-10442 and EMPIAR-11830, respectively.

Alignment and reconstruction settings

All series were aligned at bin 8, with eight times the input pixel size. Coarse two-dimensional alignment supplied the starting image translations. The Greco optimizer started from seven central views for odd-sized series and six for even-sized series. Local image patches were 33 × 33 pixels, and local reference volumes were 33^3 voxels. Patch sizes, search radii and fit tolerances are given at this bin-8 sampling.

The angular correction K used regularization 0.05 and a gain limit of 4. The spherical mask m had a radius of 14 voxels, with a cosine taper over its outer 3 voxels. When a new view enters, an initial search uses a radius of 32 pixels. A Gaussian prior with $\sigma = 8$ pixels

favours corrections near the predicted position. Generalized cross-correlation used an exponent of 0.25 and a 4-pixel search radius. When a proposed fit exceeded the allowed shift range, a rescue pass repeated the measurements with a 16-pixel radius. After an accepted rescue, refinement returned to the 4-pixel search. The joint fit used a soft-L1 scale of 0.2 pixels to reduce the influence of large residuals.

One pass rebuilt the local references, measured image corrections and updated the joint geometry. A stage converged when the median change in image position was below 0.10 pixels and its 90th percentile was below 0.25 pixels for three consecutive passes. Normal and rescue attempts shared a limit of 60 passes per stage. Smoothness was disabled except for DS-10442, where it was set to 4. This setting controls the additional neighbouring-view constraints described in the Approach. Each initial alignment run included one extra round that rebuilt the landmarks and refined the accepted angular range. The core fit kept the supplied tilt angles and axis orientation fixed.

Final tomograms were reconstructed by weighted backprojection using the accepted views and their fitted geometry. For this step, Greco returned to the motion-corrected sums at native sampling, applied spatial Wiener CTF correction and resampled the images at the final binning in Table 1. The reliability filter was used only for alignment. The 50% thickness setting retained the central half of the reconstruction grid along z. Saved coordinate transforms map particle positions from the exported tomogram back to the tilt images.

Additional geometry refinement

Geometry refinement starts from the accepted alignment. It can use the reconstruction landmarks directly, or replace them with particle positions from STA. Local reference volumes are reconstructed from the measured tilt-image patches, with the target view excluded. The fit updates the rotation and translation of each view together with the three-dimensional landmark positions. Each landmark has one position shared across the views. Proposed updates are checked against the measured image patches before acceptance. The tomogram is then reconstructed using the refined geometry.

TinyCARE denoising

TinyCARE uses the noisy-pair training principle of Noise2Noise and cryoCARE (Buchholz et al., 2019; Lehtinen et al., 2018). Movies are split into two disjoint frame sets, and both half tomograms are reconstructed with the same geometry. A patch from one half is used as input and the corresponding patch from the other half is used as the training target. The network learns to reduce their mean squared difference. Both halves share the specimen signal, while their noise is partly independent. This allows training without a clean reference. Shared image processing means that noise independence is an approximation.

TinyCARE uses a compact residual three-dimensional U-Net for fast per-tomogram training. It has 140,509 trainable parameters, two downsampling stages and 12 initial channels. It uses 3^3 convolutions, ReLU activations, max pooling and nearest-neighbour upsampling. The output convolution predicts a correction that is added to the input. The small network and mixed-precision computation reduce the processing cost. The training principle itself is the same as in established noisy-pair methods.

Training used patches of up to 72^3 voxels at 1,200 positions: 1,080 for training and 120 for validation in separate regions along y. Augmentation used quarter-turn rotations around y and exchanges of the two halves. Training ran for 3,000 successful updates over 15 epochs of 200 steps, with batch size 16 and an initial Adam learning rate of 4×10^{-4}. The final

checkpoint was used. Training and prediction used FP16, while model parameters, optimizer state and loss accumulation remained in FP32. Dynamic loss scaling handled overflows. Prediction used overlapping tiles with a reflection halo. Each half tomogram was denoised separately, and the two outputs were averaged. Denoised maps were used for inspection and segmentation; STA used the underlying particle images.

For Figure 6, weights were saved at the labelled steps during one continuous training run per method. Both methods used the same half tomograms, 72^3 patches, batch size 16, seed 17, the same initial learning rate and their native 90/10 spatial training/validation split. cryoCARE 0.3.0 used TensorFlow 2.15.1, a depth-three U-Net with 16 initial channels, and FP32. Its default schedule was 100 epochs of 200 steps. Outputs were stored as FP16. The displayed region lies in the training portion of the volume. All panels use one linear grayscale window, from the pooled 0.5th and 99.5th percentiles of the eight denoised regions, without a spatial display filter or local contrast adjustment. The deposited reference in Figure 4a was processed externally and is not a TinyCARE output.

The L40S timing test repeated TinyCARE to 3,000 steps and cryoCARE to 20,000 steps on the same input pair. Training times include validation, and prediction times cover the full volume. Each method was measured once. Queue waiting, separate input preparation and separate output storage were excluded; totals were calculated before rounding. Figure 6 uses the original checkpoint images, while the separate L40S test provides the reported times.

Particle picking

Greco Picker locates candidate particles by three-dimensional template matching, following the correlation and angular-sampling principles used in PyTom (Chaillet et al., 2023; Hrabe et al., 2012). The template and its mask are resampled to the tomogram pixel size. Template contrast and handedness are set to match the reconstructed volume. Each rotated template is compared with local density, and the highest correlation and corresponding orientation are retained at each position. Normalization within the mask reduces sensitivity to local changes in mean density and contrast. Fourier weights account for the accepted tilt angles and CTF.

To estimate local background matches, the search uses a phase-randomized template in both contrast signs. Their maximum score maps are averaged and subtracted from the original score map. The spatial mean of this background estimate is then added back. The automatic threshold uses the score variation and a Gaussian background approximation, accounting for the number of tested positions and orientations (Rickgauer et al., 2017). A user-selected factor scales this threshold. Peaks above it are selected in descending score order, with a minimum separation and boundary exclusion set by particle size.

Exhaustive mode evaluates the full orientation set across the volume. Fast mode first screens the volume with a smaller, well-spaced orientation set, then tests the full set at candidate positions and nearby voxels. Both modes process orientations in GPU batches and reuse shared calculations. All picking runs reported here used tomograms before denoising, Exhaustive mode with 10° angular sampling (15,552 orientations), and a 400 Å high-pass filter.

Subtomogram averaging

Greco includes a minimal STA implementation whose only purpose here is to evaluate the quality of reconstructions produced with GRECO. It contains adapted RELION routines and follows the tilt-image processing model developed in RELION (Burt et al., 2024; Zivanov et al., 2022). Small particle images are extracted from the motion-corrected tilt series. The

tomogram provides the particle locations; refinement uses their measured two-dimensional views. Each predicted view includes the CTF at the particle's position and depth.

Three-dimensional classification groups particles while keeping their positions and orientations fixed. Refine3D adjusts particle positions and orientations in continuous local steps. A trust-region optimizer controls the step size and accepts a change when it improves agreement with the images. Particles keep their assignment to two half sets, and each particle is aligned against the reference from its own half. Finer spatial frequencies enter the alignment as agreement between the half maps improves.

CTF refinement jointly adjusts defocus, astigmatism and contrast scale for each view. Polishing adjusts the tilt-series geometry and models particle motion during acquisition. Nearby particles are encouraged to move in similar ways, which helps constrain weak individual measurements. Particle motion is fitted within each half set; the tilt geometry and optical corrections are shared between the halves. Proposed corrections are checked against the measured particle images before they are retained.

Resolution was assessed by Fourier shell correlation (FSC), which measures agreement between the two half maps at different spatial frequencies. The FSC analysis used the 0.143 criterion and phase-randomization correction to account for correlations that the mask introduces (Chen et al., 2013; Rosenthal & Henderson, 2003). Postprocessing combines the half maps and applies sharpening and low-pass filtering according to the supported resolution.

Segmentation and structural context

The segmentation workflow combines the reconstructed density with membrane surfaces and particle annotations.

Membranes were segmented from TinyCARE-denoised tomograms using the pretrained MemBrain-seg model associated with MemBrain v2 (Lamm et al., 2026).

Particle annotations are built by placing the refined average at each particle's saved position and orientation. Several particle types can be displayed together with the membrane surfaces and a tomogram section. This shows their spatial organization within the specimen. Each placed particle is a transformed copy of its class average, so the layer describes molecular positions and orientations, not the density resolved independently for each molecule.

Discussion

GRECO uses local structure in the specimen to determine one consistent reconstruction geometry. Each image translation is supported by many landmarks, and each landmark is constrained by several views. This coupling helps prevent one strong local match from determining an image shift on its own. Overlapping angular tracks connect these constraints across the series, even when individual landmarks are useful over only part of the angular range. The strength of the approach comes from combining local measurements that must agree with the same three-dimensional arrangement.

Starting from the central views makes the first registration steps possible before landmark depth is well determined. Near zero tilt, a depth error produces a smaller displacement in the image, so provisional landmarks can still guide local matching. As more views are accepted, the wider angular range adds information about depth and the landmark cloud is rebuilt. The structural saliency score identifies useful local density, while columns help distribute candidates across the image plane when viewed from different directions. Alignment and landmark localization therefore develop together as the reconstruction gains information.

The examples show how this strategy uses different sources of contrast. In the ER-derived vesicles, many landmarks coincide with gold fiducials, while others contain specimen density. In the CZII mixture, biological structures provide the landmarks without added fiducials. Their molecular identity is not needed during alignment. This allows the same selection strategy to use whatever local structure provides useful positional information, across samples with different compositions and preparation methods.

The cellular examples extend this use to intact near-minimal cells and in situ lamellae. The lamella reconstructions supported both ribosome averaging and segmentation of membranes with different shapes. The minimal STA implementation provides a downstream assessment of the structural information available after processing with Greco. Particle positions obtained by STA can also guide further geometry refinement. Together with membrane segmentation, this connects molecular analysis with the organization of the surrounding cell and gives a practical use to the reconstructed volumes beyond visual inspection.

The near-minimal cell also shows where additional constraints can help. Landmark fitting alone could not extend the alignment beyond the central seed, whereas adding measured shifts between neighbouring views allowed the full series to be reconstructed. These image relationships complement the landmark measurements when local information is weak. Sufficient contrast and overlapping angular tracks remain necessary for the shared fit. The core optimizer also relies on the supplied tilt angles and axis orientation; the optional geometry refinement provides a later opportunity to adjust the viewing directions.

The implementation is fast across preprocessing, registration, reconstruction and denoising. GPU computation and batched operations reduce the cost of processing the movies and fitting the geometry. Local matching works with small reference volumes, while the final tomogram is reconstructed at the requested sampling. TinyCARE adds a compact network with short training and prediction times. In the examples shown here, 3,000 training steps provided a useful balance between visible detail and processing time. This makes denoising a practical part of the workflow for inspecting the specimen and preparing membrane segmentation.

This efficiency provides a practical basis for live processing during data collection. A natural next step is to process newly acquired tilt series as they become available, producing reconstructed and denoised volumes while acquisition continues. This would give early feedback on specimen content and alignment quality and could help guide the selection of further acquisition targets. Bringing these steps together would allow users to assess their data during the experiment and make better use of the available microscope time.

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Declaration of interests

The author declares no competing interests.

Naming convention

GRECO stands for *Geometric Registration of Electron Cryotomography Observations* and names the mathematical algorithm described in this work. Greco is the software that implements this algorithm and provides the surrounding processing workflow. Within this software, the Greco optimizer is the component that performs the registration and geometry optimization. The lowercase name `greco` refers to the executable used to launch the software and to the repository name.

Code availability

Greco is available at github.com/akribeia/greco under the GNU Affero General Public License, version 3 (AGPLv3), with the applicable notices for third-party components. The repository also includes the original MATLAB experiment.

Data availability

The input data are publicly available. The ER-derived vesicle movies are deposited in DataverseNL at doi:10.34894/OLYEFI. The CZII challenge data are available from the CryoET Data Portal as DS-10440. The minimal-cell example is DS-10442, run RN-16210, which also provides the denoised reference tomogram and annotations. The *C. reinhardtii* data are available as EMPIAR-11830.

AI declaration

OpenAI Codex assisted with code development, testing, literature searches, manuscript writing and editing, data analysis, and figure preparation. The author reviewed and revised all AI-assisted work and takes full responsibility for the final manuscript, software, and results.

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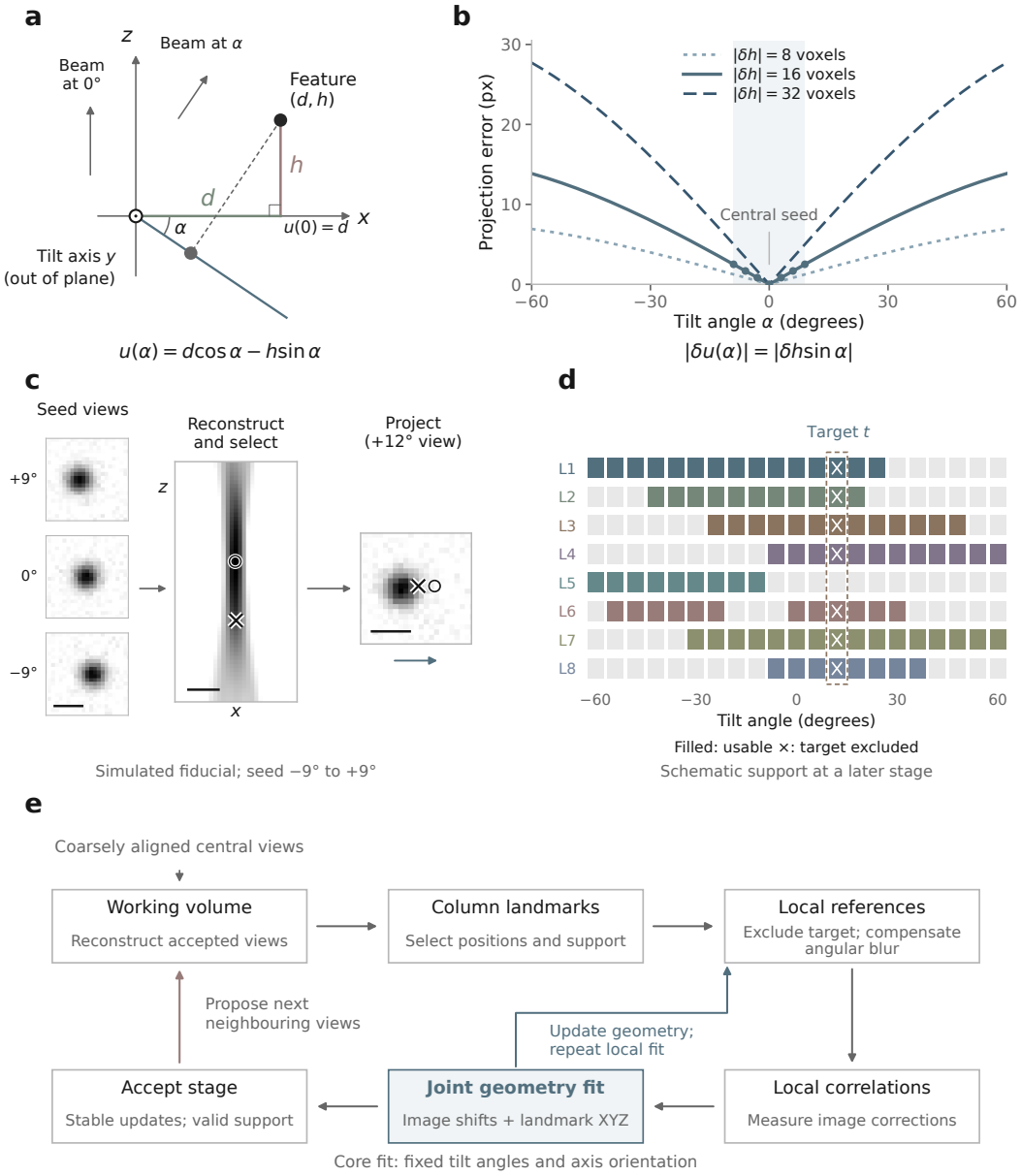


Figure 1

Geometric motivation and workflow of the Greco optimizer. *(a) A specimen feature at (d, h) in an x–z section. The tilt axis is perpendicular to the page. Rotating the viewing direction gives the projection convention in Equation 1. (b) Image prediction error caused by depth errors of 8, 16 and 32 voxels, with the lateral position held fixed and equal voxel and detector sampling. The shaded range and dots indicate an example central seed. Near zero tilt, the projections are less sensitive to depth uncertainty. (c) A simulated Gaussian fiducial in seven noisy views from −9° to +9° (+9°, 0° and −9° shown), its weighted-backprojection reconstruction in an x–z slice, and the selected point projected into an independent +12° image. The seed elongates the feature along the beam axis at zero tilt. The structural saliency score selects a provisional point within one column: its depth differs by 14.5 voxels from the true centre, but its horizontal prediction error at +12° is only 3.2 pixels. The open circle in the reconstruction marks the true position; crosses mark the selected point and its projection. The blue arrow indicates the image shift that brings the observed feature into alignment with the projected landmark. Scale bars: 8 pixels or voxels. (d) A schematic support matrix for eight landmarks at a later stage, separate from the single-fiducial simulation. Each landmark has a different colour. Filled cells are usable observations; crosses mark the target excluded from its own reference. Pale cells are unused. (e) Local references, correlations and the joint geometry fit are repeated within a stage. Accepted geometry allows neighbouring views to enter and the working volume and landmarks to be rebuilt. Tilt angles and the axis orientation remain fixed in the core fit.*

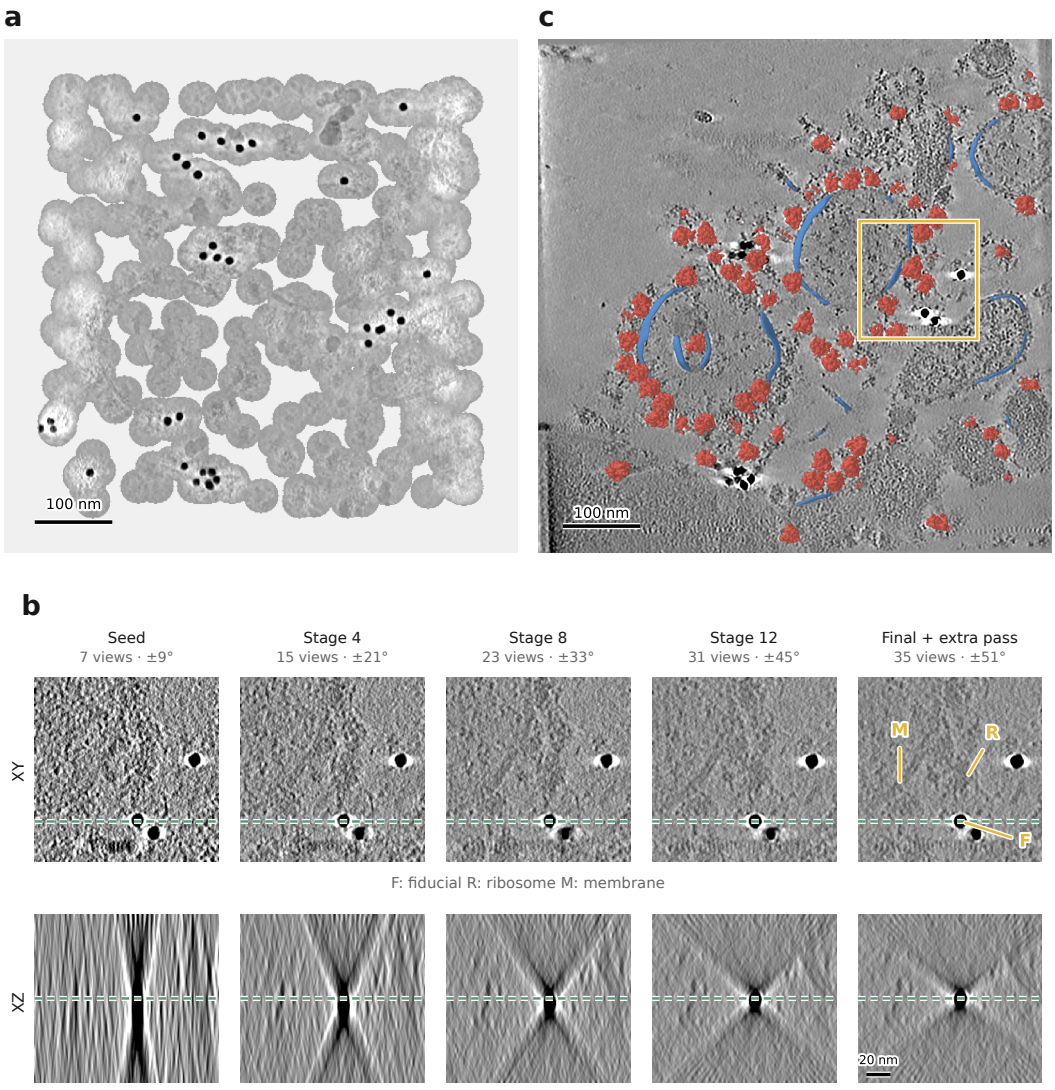


Figure 2

Landmarks and local structure in ER-derived vesicles. *(a) Final landmark projection at 0° tilt. Many selected regions contain gold fiducials; others contain specimen density. (b) Matched XY and XZ sections at five stages of the Greco optimizer: the seed, stages 4, 8 and 12, and the final extra pass. Labels give the number of views and the angular range. F, R and M indicate a fiducial, a ribosome and a membrane. Dashed lines locate the intersecting sections. The XZ row averages three adjacent Y slices. These weighted-backprojection sections use the same Wiener-filtered input, voxel size (6.896 Å) and linear display window. A mild Gaussian filter (σ = 0.6 pixels) is applied for display. Scale bar: 20 nm, common to all panels in (b). (c) Front view of a TinyCARE slice, with illuminated membrane surfaces (blue) and placed ribosome averages (red). The 78 ribosomes are shown at their refined positions and orientations. The complete particle layer contains 169 ribosomes retained after classification and refined to a reported FSC resolution of 29.3 Å. The view is orthographic, so the box marks the same XY region as (b). Scale bars in (a, c): 100 nm.*

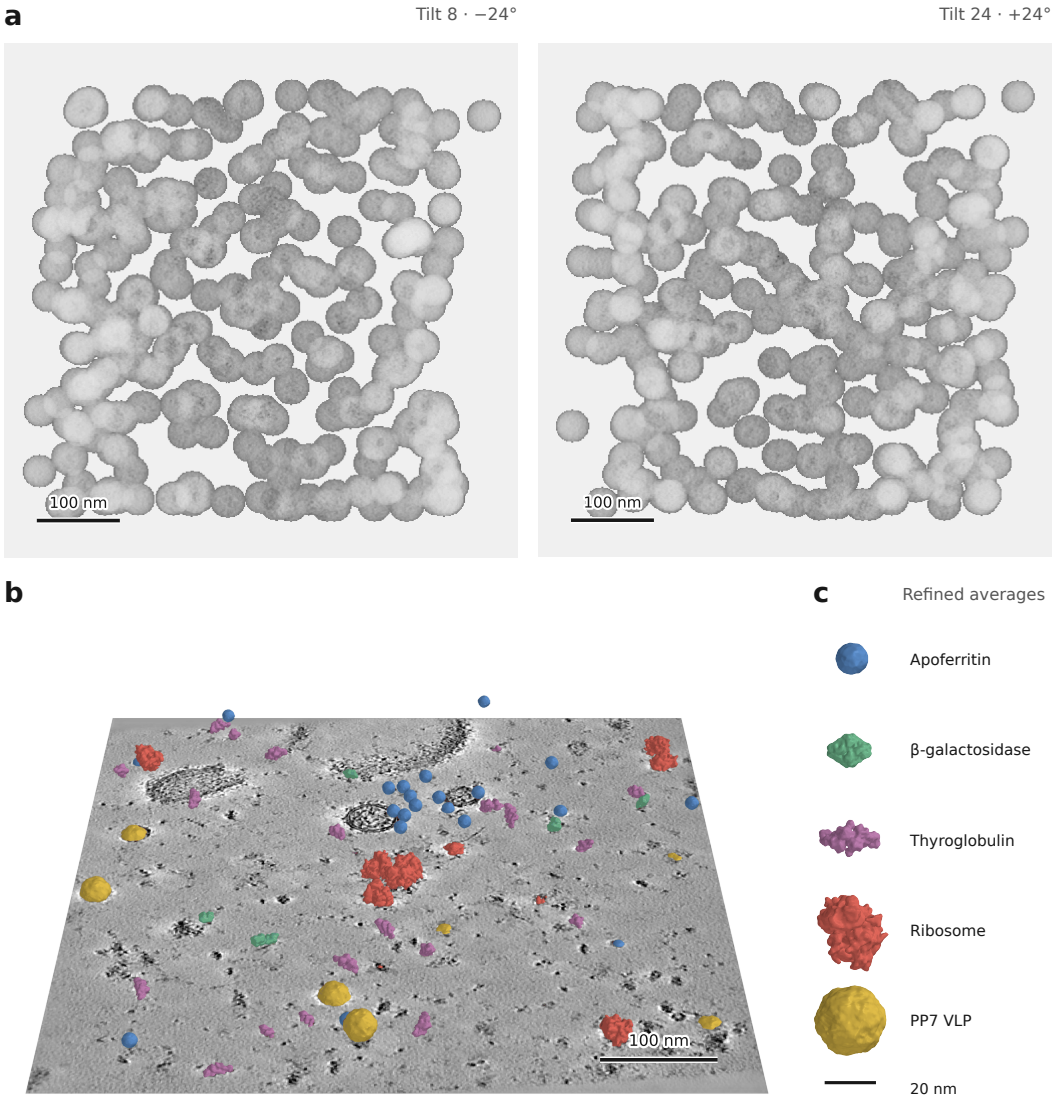


Figure 3

Landmarks and structural annotation in the CZII challenge specimen. *(a) Projected density around the final landmark positions at −24° and +24°, using the same global contrast curve, which gently compresses extreme intensities to make local features easier to see. Of the 1,080 selected three-dimensional landmarks, 694 had support at −24° and 687 at +24°. Regions around these landmarks overlap, so the two projections do not show a separate visible feature for every landmark. Their overlapping angular tracks link measurements across views and help constrain a consistent geometry. Scale bars: 100 nm. (b) Perspective view of a TinyCARE slice after geometry refinement guided by ribosome positions from STA. Coloured surfaces show 69 placed particles: 21 apoferritin particles, 8 β-galactosidase particles, 22 thyroglobulin particles, 11 ribosomes and 7 PP7 VLPs. Depth is not exaggerated. The 100 nm bar lies in the slice plane. (c) Final refined averages for the five particle types, at a common physical scale (20 nm bar). Colours match (b). From top to bottom, the reported Refine3D FSC resolutions are 20.3 Å for apoferritin, 11.8 Å for β-galactosidase, 13.9 Å for thyroglobulin, 20.9 Å for ribosome and 22.4 Å for PP7 VLP (phase-randomization-corrected FSC). The averages and their placements come directly from STA of the seven CZII tomograms. Each annotation in (b) places the corresponding average at a particle position and orientation refined from these data; it does not represent the density resolved independently for that individual molecule.*

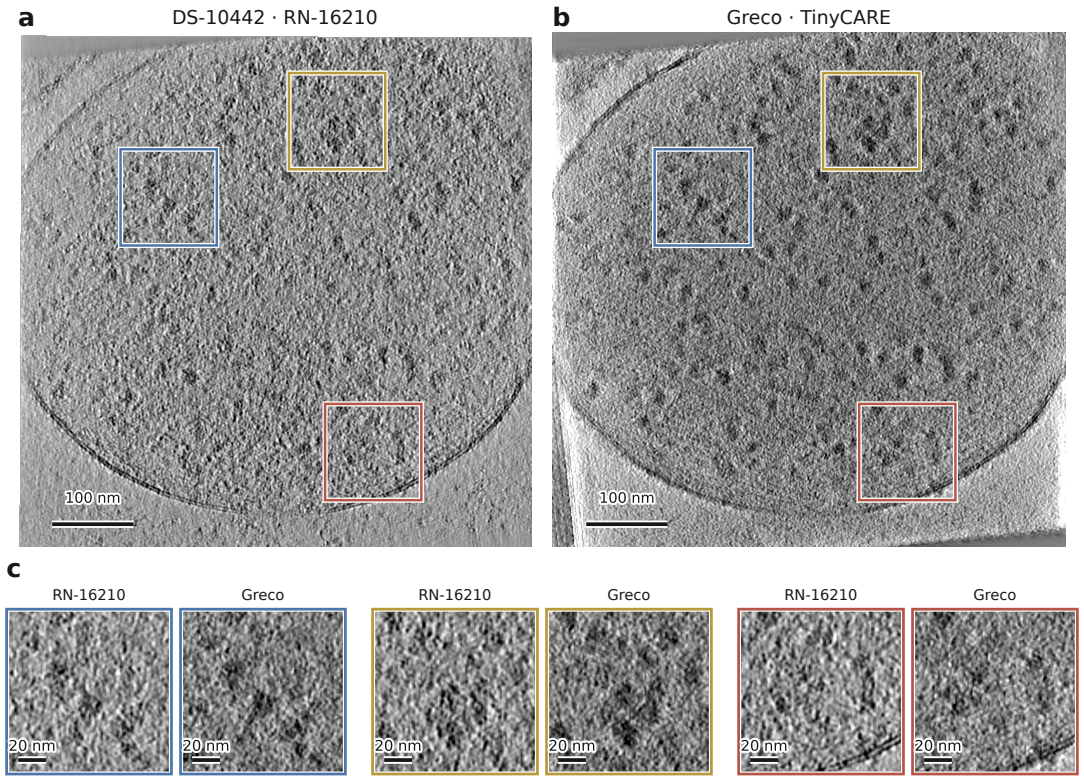


Figure 4

An intact near-minimal cell reconstructed with Greco. *(a) Deposited denoised tomogram of plunge-frozen* Mycoplasma mycoides *JCVI-Syn3A from the CryoET Data Portal, DS-10442, RN-16210, deposited tomogram 100 (Paraan et al., 2024). (b) Greco reconstruction with Smoothness set to 4 and TinyCARE denoising. All 31 views contributed to this reconstruction. The run without Smoothness stopped at the seven-view seed and could not extend the alignment. The overviews show a corresponding physical plane after global rigid registration and correction of the Y-axis convention. Both are displayed at 12.32 Å per pixel. Scale bars: 100 nm. (c) Three pairs of enlarged regions, marked by matching coloured boxes and borders: two regions of crowded cell interior and one near the cell boundary. The deposited RN-16210 map is on the left and the Greco result on the right in each pair. Patches cover the same coordinates and use the display window of their overview. Scale bars: 20 nm. The comparison illustrates cellular detail in separately processed maps, rather than measuring denoising accuracy. The visible differences reflect the combined effects of alignment, reconstruction and denoising.*

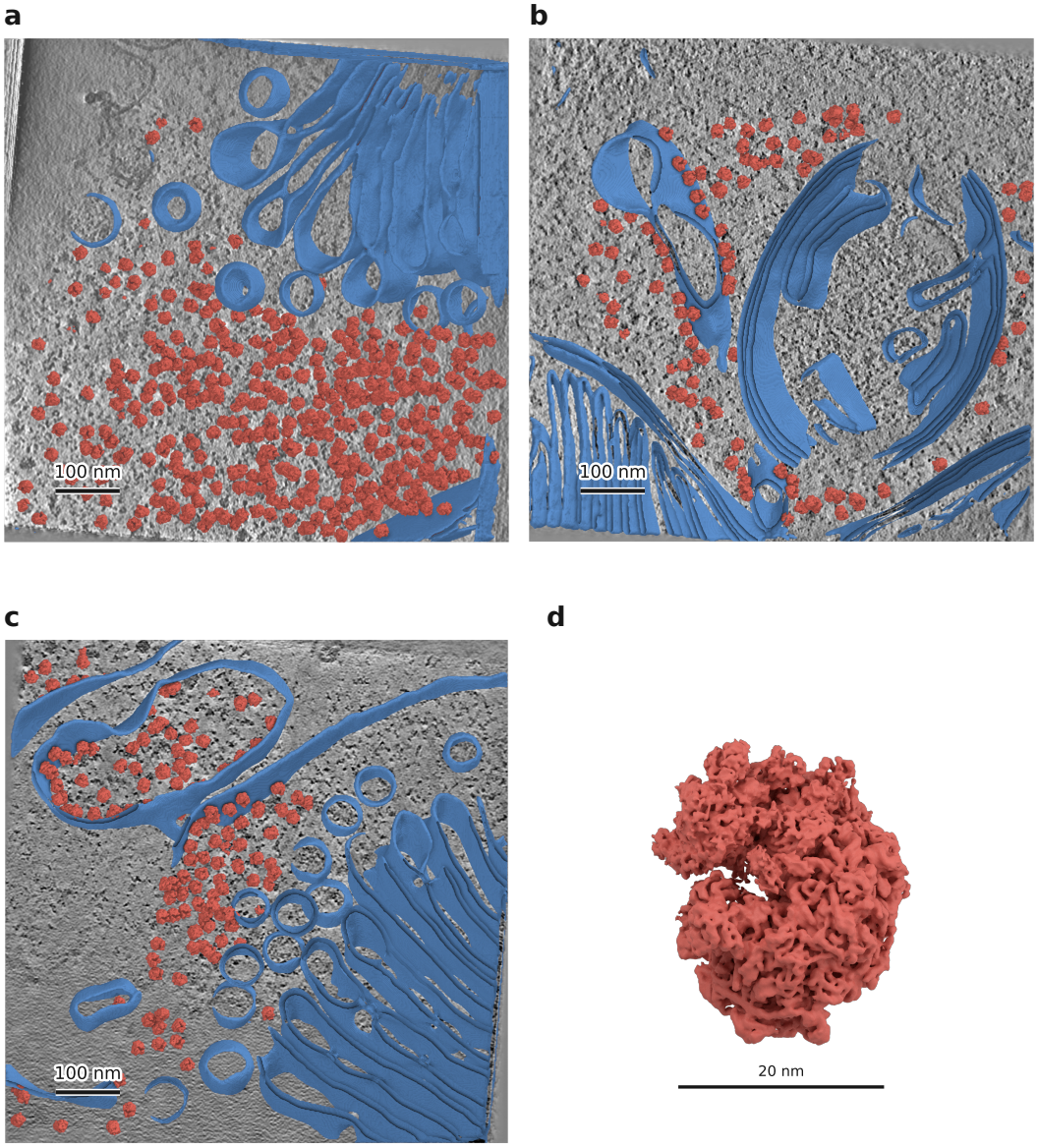


Figure 5

In situ cellular context and ribosome averaging in EMPIAR-11830. *(a–c) Frontal views of three* C. reinhardtii *lamella tomograms after geometry refinement guided by ribosome positions from STA and TinyCARE denoising. Red surfaces show 414, 111 and 149 placed ribosomes, respectively, at their refined positions and orientations. Blue surfaces show segmented membranes. Scale bars: 100 nm. (d) Whole-ribosome map refined from 11,976 particles using a broad mask. Its reported phase-randomization-corrected FSC resolution is 6.36 Å. Scale bar: 20 nm. A separate large-subunit-masked refinement reached 5.38 Å; that map is not shown in (d).*

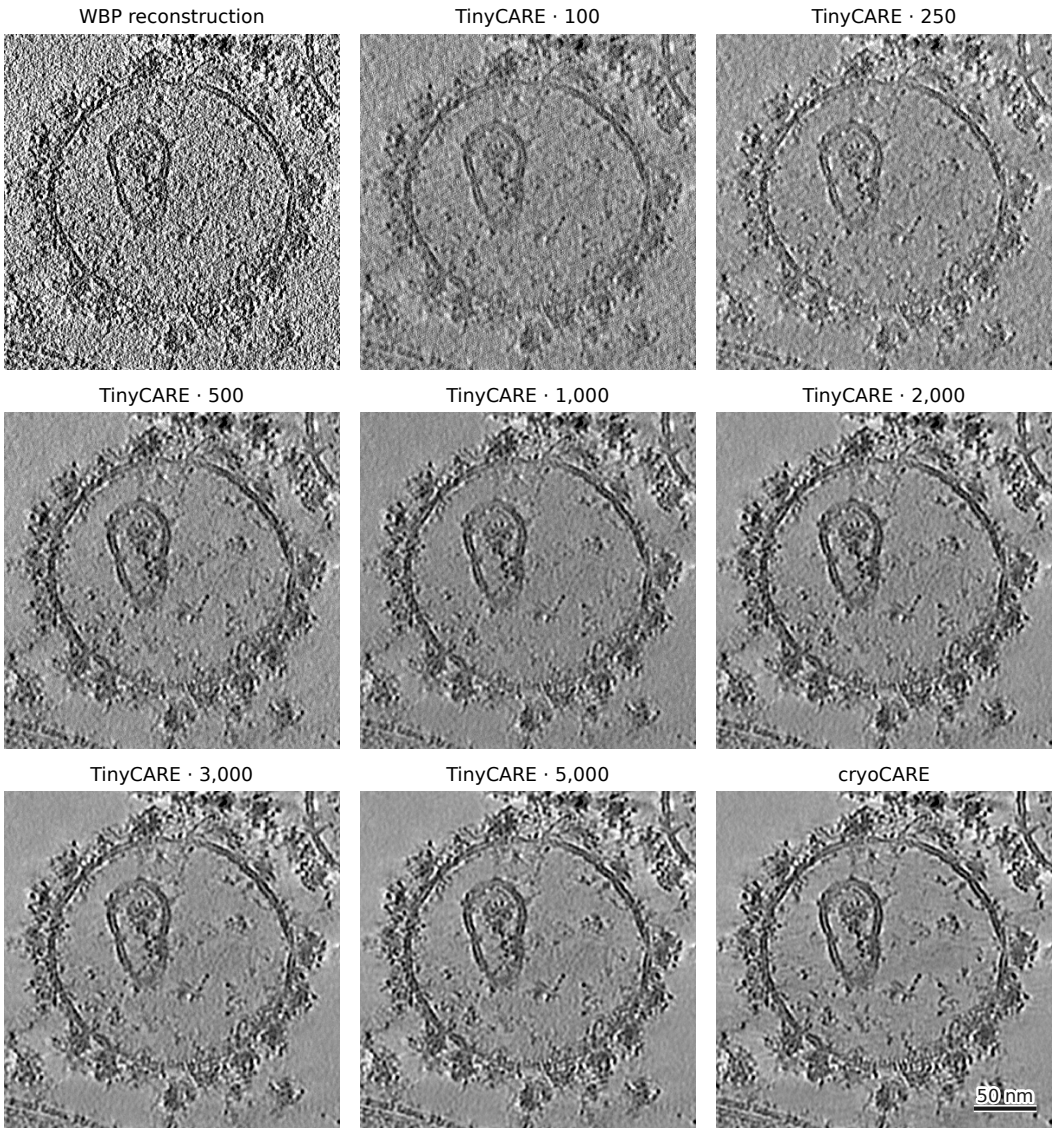


Figure 6

TinyCARE training progression compared with cryoCARE in the ER-derived vesicle tomogram. *From left to right and top to bottom: the WBP reconstruction before denoising; TinyCARE after 100, 250, 500, 1,000, 2,000, 3,000 and 5,000 training steps; and cryoCARE after its default 20,000 steps. The WBP panel is the mean of the two input half reconstructions. All panels show the same region with a common linear grayscale window. The displayed region was included in training. Scale bar: 50 nm (lower-right panel). Sampling: 6.896 Å per pixel.*

Appendix I
Recovering the tilt-axis orientation from central views

The central seed can also provide a starting point when the tilt-axis orientation is inaccurate. Near zero tilt, even a large error in axis orientation can produce small projection errors for some features. A seed reconstructed with the wrong axis may therefore still contain local structures whose projections are useful for registration. These structures can guide a joint correction of the axis and landmark positions. In this setting, axis estimation starts during the seed fit and continues as views are added.

Figure Ia adds the axis orientation β to the geometry of Figure 1a. At zero tilt, h measures depth along the beam, d measures distance across the axis, and q measures position along it. Here, h is the feature's depth; δh in the Approach describes an error in its estimated depth. The axis is assumed to lie in the image plane, and β describes its deviation from the image y direction. The equations in Figure Ia give the two image coordinates of a feature after tilting by α. Coordinates are centred on the rotation origin, with distances expressed in pixels at the working sampling.

To compare a true and a tested axis, the physical point and the image coordinate frame must stay fixed. Let $\delta = \beta_{\text{tested}} - \beta_{\text{true}}$ be the axis error, and choose coordinates with the true axis along y. The point then has coordinates (d, q, h) in the (x, y, z) frame. Its coordinates along and across the tested axis change when δ changes. Let D be the distance between its true and tested projections. The projection model gives the bound

$$D \leq (1 - \cos\alpha)|\sin\delta|\sqrt{d^2 + q^2} + 2\left|h \sin\alpha \sin\left(\frac{\delta}{2}\right)\right|. \tag{I.1}$$

This expression separates the effect of position in the zero-tilt plane from the effect of depth. The first term is the exact error when $h = 0$; the second is the exact error when $d = q = 0$. For a general point, these two displacement vectors can partly reinforce or cancel each other. Their magnitudes give the bound above. It remains valid for a large axis error, including $\delta = 90°$.

Near zero tilt, the first contribution decreases as α^2, while the depth contribution decreases as $|\alpha|$, with α in radians. Figure Ib shows this behaviour within the central seed, from $-9°$ to $+9°$. Even when the true and tested axes are perpendicular ($\delta = 90°$), the illustrated points with $h = 0$ remain below 2 pixels of projection error throughout this interval. A large axis error therefore does not displace all features equally: some near the zero-tilt plane can still provide useful local references. Smaller distances across or along the axis also help.

When the first incoming view is at $\alpha = +12°$ and the axis error is $\delta = 90°$, a point with $d = 100$ pixels and $q = h = 0$ has a projection error of 2.19 pixels. A point with $q = 100$ pixels and $d = h = 0$ gives the same error. For a point at $h = 100$ pixels, with $d = q = 0$, the error is 29.40 pixels. Figure Ic illustrates this difference across points distributed in three dimensions at the same incoming tilt angle.

The same geometry explains what happens during reconstruction. For an ideal orthographic projection, each image point defines a backprojection ray. Using the wrong axis changes this ray. The shortest distance from the true three-dimensional point to the wrong ray equals D, when image and volume sampling are equal. Small projection errors therefore also mean that the rays pass close to the feature. Their overlap can preserve a local density even if its shape and depth remain inaccurate. This gives the first landmark candidates something to follow.

At exactly zero tilt, $D = 0$ for any axis orientation, so that image alone cannot determine the axis. Useful information comes from several nonzero central views and features distributed through the specimen. Figure Ic shows why only part of the landmark cloud may be needed to begin this process. The physical points and their true projections stay fixed while the tested axis changes. Even at $\delta = 90°$, three of the 24 off-origin points lie within the ordinary 4-pixel search range (green). Their small projection errors make them possible starting points for local matching, provided their image patches contain useful structure.

A small image error can therefore coexist with a large axis error. For these points, bringing the predicted and observed positions together may require only a few pixels of local correction while the common axis changes by many degrees. GRECO can use such a subset to guide the rotation before most of the cloud becomes easy to match. The information comes from the pattern of corrections across their tracks, linked through one geometry. The point at the origin has zero projection error for every axis orientation and supplies no angular information by itself.

The common-axis fit connects the local displacement measurements through one angular correction. Following the displacement convention of Equation 4, its local prediction becomes

$$\boldsymbol{v}_{k,t} \approx \Delta \boldsymbol{s}_t - J_t \Delta \boldsymbol{r}_k + \boldsymbol{g}_{k,t} \Delta\beta. \tag{I.2}$$

Here, $\boldsymbol{g}_{k,t}$ is the change in the predicted image correction per unit change of the axis angle, evaluated at the current projected landmark position. The increment $\Delta\beta$ is shared by all supported observations in one common-axis update. Greco fits this first-order approximation with the robust loss and translation constraint used in the main joint fit. The axis and the three-dimensional landmark positions change together. Their projected motions can largely compensate each other, leaving a local image correction of only a few pixels.

This behaviour also appears in the recorded fits. During the ER-vesicle seed fit, one update changed the common axis by 32.5°, while the measured local corrections had an RMS of 0.87 working pixels per image coordinate. After applying an update, Greco repeats the local correlations and recalculates the geometric relation. The angular correction comes from the joint fit, without evaluating a grid of candidate axis orientations.

Figure Ic also illustrates how this process can gain support. As the axis error decreases from 90° to 45°, 20° and 10°, the number of green points increases from 3 to 4, 8 and 22. More of the specimen can then supply local measurements within the same search range. In GRECO, rebuilding the working volume and selecting landmarks again allows this improved geometry to bring further useful observations into the fit. The added tracks connect more of the specimen and angular range, helping to constrain the following updates.

The recovered axis therefore emerges from the geometry shared by the images and landmark tracks. Each image has one translation, each landmark has one three-dimensional position, and the common axis must explain their relative projected motions across views. A usable subset can initiate the correction; a broader set can then help stabilize it. GRECO selects candidates by structural saliency and supporting observations, so the exact sequence of useful landmarks depends on the specimen and its images.

After the common-axis fit, Greco can allow a separate axis correction $\Delta\beta_t$ for each view. These view-specific parameters still enter one joint landmark fit. A following phase can also refine the acquisition tilt angles α_t, while keeping the central tilt angle anchored and preserving their order. The resulting model refines image translations in x and y, the in-plane axis orientation β_t and the tilt angle α_t for each view. Each view can then better follow its measured landmark positions while sharing the same three-dimensional cloud. The recovery test below isolates the common-axis stage.

The test included 36 tilt series from the four datasets used in this work. All reported runs started with an axis orientation of 0°, approximately a quarter-turn from the declared orientations. This deliberately large mismatch provides a demanding test of axis recovery in Greco. CTF reliability weighting used a global defocus estimate for each view, without a spatial gradient based on the declared axis.

Table I summarizes the recovered orientations across the four datasets.
These results show recovery from a large initial axis error across different specimen types. They support the use of central views and distributed landmarks to estimate the common orientation during alignment. They do not establish the absolute angular accuracy or determine specimen handedness.

Table I. Common-axis recovery from a o° initial orientation.

Dataset	Initial (°)	Declared (°)	Recovered (°)	Difference (°)
ER-derived vesicles (1 series)	0	−88.70	−89.70	−1.00
CZII challenge (7 series)	0	−96.00	−96.41 to −95.64	−0.41 to +0.36
DS-10442, RN-16210 (1 series)	0	+83.94	+84.13	+0.19
EMPIAR-11830 (27 series)	0	−95.00	−96.92 to −95.65	−1.92 to −0.65

All angles are in degrees. Difference = recovered − declared. The declared axis is the nominal acquisition value and should not be treated as ground truth.

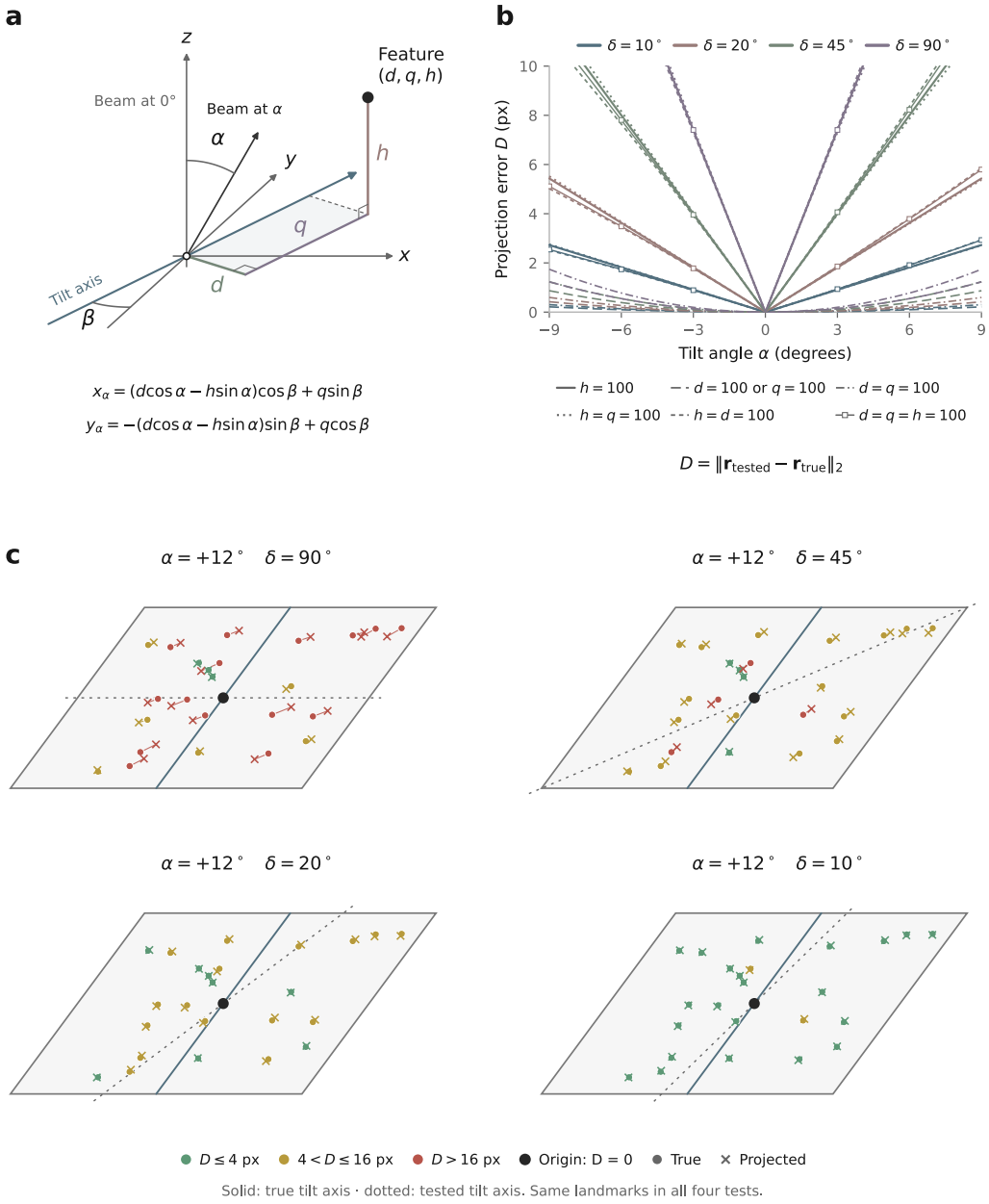


Figure I

Geometry of axis recovery from central views. a, *A feature has transverse position d, position q along the tilt axis and depth h at zero tilt. The angle β sets the axis orientation in the image plane, and α is the specimen tilt angle.* ***b,*** *Exact projection error D within the central seed ($-9°$ to $+9°$) for the indicated positions and axis errors δ. Even with a $90°$ axis error, the curves with $h = 0$ remain below 2 pixels throughout this interval. Positions are in pixels; coordinates not listed in the legend are zero. The cases with only $d = 100$ or only $q = 100$ have identical error magnitudes. The vertical range is limited to 10 pixels.* ***c,*** *True projections (circles) and tested projections (crosses) of the same synthetic points at $\alpha = 12°$, for axis errors δ of $90°$, $45°$, $20°$ and $10°$. Colours indicate the projection error: green, $D \leq 4$ pixels; yellow, $4 < D \leq 16$ pixels; red, $D > 16$ pixels. The number of green points increases from 3 to 4, 8 and 22 as the axis improves, illustrating how more features can enter the same local search range. The origin has $D = 0$ for every axis orientation. Solid and dotted lines indicate the true and tested axes. These panels compare fixed geometries; they are not successive iterations of a fitted trajectory.*